

§ Group

Examples

- Permutation / symmetric group S_X , X : non-empty set.

$$S_X := \left\{ \text{bijections } X \xrightarrow[1:1]{\sigma} X \right\}$$

Multiplication = $\circ$, unit = id_X , σ^{-1} = inverse map of σ .

- Alternating group / 交错群 $A_n := \{\sigma \in S_n \mid \text{sgn}(\sigma) = 1\} \subset S_n$: subgroup.
- General linear group / 一般线性群

Let V : F -vector space, $\text{GL}(V) := \{\text{isomorphisms } V \xrightarrow{\sim} V\}$

- Special linear group / 特殊线性群

$\dim V < \infty$: $\text{SL}(V) := \{T \in \text{GL}(V) \mid \det T = 1\} \subset \text{GL}(V)$: subgroup.

- Orthogonal group / 正交群

Let V : $\text{IPS}/_{\mathbb{R}}$, $\dim < \infty$. $\text{O}(V) := \{T : V \xrightarrow{\sim} V \text{ as } \text{IPS}/_{\mathbb{R}}\}$.

$\text{SO}(V) := \text{O}(V) \cap \text{SL}(V)$.

- Unitary group / 酉群

The $\mathbb{C}$ version of orthogonal group. Includes $\text{U}(V)$, $\text{SU}(V)$.

- Variants for matrices:

$\text{GL}(n, F) := \{A \in M_{n \times n}(F) \mid \text{invertible}\}$,

$\text{SL}(n, F) := \{A \in \text{GL}(n, F) \mid \det A = 1\} \subset \text{GL}(n, F)$,

$\text{O}(n) := \{A \in \text{GL}(n, F) \mid {}^t A A = 1_{n \times n}\}$,

$\text{SO}(n) := \text{SL}(n, \mathbb{R}) \cap \text{O}(n)$,

$\text{U}(n)$, $\text{SU}(n)$ are similar.

- $(\mathbb{Z}, +)$ with unit = 0, $(\mathbb{Z}/n\mathbb{Z}, +)$ with unit = the class $n\mathbb{Z}$.

Note that both groups are abelian.

In general $(R, +, \cdot) : \text{ring} \implies (R, +) : \text{group with unit} = 0_R$.

- R ring $\implies R^\times$ is a group w.r.t. multiplication, unit = 1_R .

Notation G group, $S \subset G$ as subset,

$$\langle S \rangle := \{s_1^{a_1} \dots s_n^{a_n} \mid n \in \mathbb{Z}_{\geq 0}, s_i \in S, a_i \in \mathbb{Z}\} \subset G$$

as subgroup, and is the smallest subgroup of G containing S .

We say S is a family of generators of G , iff $\langle S \rangle = G$.

Examples $\text{S}(n)$ is generated by $s_i = (i \ i+1)$, $\mathbb{Z}/n\mathbb{Z}$ is generated by $1 + n\mathbb{Z}$.

If $n > 1$, then $k + n\mathbb{Z}$ generates $\mathbb{Z}/n\mathbb{Z} \iff \gcd(k, n) = 1$.

Definition Let $(G_i)_{i \in I}$ be a family of groups. Endow $\prod_{i \in I} G_i$ with the binary operation

$$(g_i)_{i \in I} (g'_i)_{i \in I} := (g_i g'_i)_{i \in I}$$

then $\prod_{i \in I} G_i$ is a group with unit $= (1_{G_i})_{i \in I}$, $(g_i)_{i \in I}^{-1} = (g_i^{-1})_{i \in I}$.

We call $\prod_{i \in I} G_i$ the direct product of $(G_i)_{i \in I}$.

If $\forall i, G_i = G$, then we write G^I . If $I = \{1, \dots, n\}$ is finite, write $G_1 \times \dots \times G_n$.

This construction is “associative” (by removing brackets):

$$(G_1 \times G_2) \times G_3 \xrightarrow{1:1} G_1 \times G_2 \times G_3 \xrightarrow{1:1} G_1 \times (G_2 \times G_3).$$

How do we show that they are the “same” groups?

§ Homomorphisms and isomorphisms

General paradigm: homomorphism “are” structure-preserving maps.

Definition G, G' groups. A map $f : G \rightarrow G'$ is a homomorphism if

$$f(xy) = f(x)f(y), \forall x, y \in G.$$

Properties If f is a homomorphism, then

- $f(1_G) = 1_{G'}, \because f(1_G) = f(1_G 1_G) = f(1_G)f(1_G)$.
- $f(x^{-1}) = f(x)^{-1}, \because f(x^{-1})f(x) = f(1_G) = f(x)f(x^{-1})$.

Remark Homomorphism between semigroups $f(xy) = f(x)f(y)$,

between monoid $f(xy) = f(x)f(y), f(1) = 1$ (not automatic).

Examples $\text{id}_G : G \rightarrow G$; inclusion $H \subset G$ (H subgroup); $\text{sgn} : S_n \rightarrow \{\pm 1\}$;

$\det : \text{GL}(n, F) \rightarrow F^\times$ (F field), $M_{n \times n}(F) \rightarrow F$ (homo. of monoid);

$(G_i)_{i \in I}$ family of groups, $\forall j \in I, p_j : \prod_{i \in I} G_i \rightarrow G_j$ (projection).

$G \xrightarrow{g} G' \xrightarrow{f} G'', f, g$ homomorphism $\implies$ so is fg by checking definition.

Definition Let $f : G \rightarrow G', g : G' \rightarrow G$ be homomorphisms. If $gf = \text{id}_G$, we say g is a left inverse of f . Right inverse is similar. We say they are mutually inverse if $fg = \text{id}_{G'}, gf = \text{id}_G$. Note that g is determined by $f : g = f^{-1}$ as maps. In this case, we say f is an isomorphism, g is its inverse which is also an isomorphism.

Examples $G \xrightarrow{\text{id}_G} G$, composition of $\simeq$ is still $\simeq$.

Given G, G' , if $\exists f : G \rightarrow G'$ isomorphism, we say G and G' are isomorphic, write $G \simeq G'$.

Proposition Let $f : G \rightarrow G'$ be a bijective homomorphism, then f is $\simeq$ of groups.

Proof To check: the inverse map $f^{-1} : G' \rightarrow G$ (between sets) is a homomorphism. □

Ditto for semigroups / monoids.

Examples

- $G_1 \times (G_2 \times G_3) \simeq G_1 \times G_2 \times G_3 \simeq (G_1 \times G_2) \times G_3$.
- F field, $V : F\text{-vs.}, \dim V =: n < \infty$.

$$\mathrm{GL}(V) \simeq \mathrm{GL}(n, F), \quad T \mapsto \text{the matrix of } T,$$

after choosing some ordered basis (not unique).

- V IPS/ $\mathbb{R}$, $\dim V = n$. $\mathrm{O}(V) \simeq \mathrm{O}(n)$.
 W IPS/ $\mathbb{C}$, $\dim W = n$. $\mathrm{U}(W) \simeq \mathrm{U}(n)$.
- $V := \{\mathrm{id}, \sigma, \tau, \sigma\tau\} \subset \mathrm{S}_4$ subgroup:
 $\sigma = (2\ 1\ 3\ 4), \tau = (3\ 4\ 1\ 2)$. Then $\sigma\tau = (4\ 3\ 2\ 1) = \tau\sigma$.
- $V' := \left\{ \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}, \begin{pmatrix} -1 & \\ & 1 \end{pmatrix}, \begin{pmatrix} -1 & \\ & -1 \end{pmatrix} \right\} \subset \mathrm{GL}(2, \mathbb{Q})$ subgroup.
- G group, $g \in G$. $\mathrm{Ad}_g : G \rightarrow G, x \mapsto gxg^{-1} \implies$ isomorphism.
 $\mathrm{Ad}_g \mathrm{Ad}_h = \mathrm{Ad}_{gh}, \mathrm{Ad}_1 = \mathrm{id}$. Hence $\mathrm{Ad}_g^{-1} = \mathrm{Ad}_{g^{-1}}$.
- $\mathrm{Aut}(G) := \{\text{automorphisms of } G\}$ is a group w.r.t. composition, $1_G = \mathrm{id}$.
 $\implies g \mapsto \mathrm{Ad}_g$ is a homomorphism of groups.